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[Operating Systems]

[CSA04

Assessment Tool-1

① Suppose a modern operating System supports multiple users and Processes with shared resources like CPU, memory, and I/o devices.

a) **Types of OS**

* Multi-user OS → supports multiple users simultaneously.
* Multi-tasking OS → Runs multiple processes at the same time.
* Time-sharing OS → shares CPU fairly among users.

b) **OS structure**

users
↓
applications
↓
System calls
↓
kernel
↳ Process Management
↳ memory management
↳ File system
↳ Device drivers
↓
Hardware

c) **System Calls**

fork() → exec() → wait() → exit()

② a) **File & Device System Calls**

* **File** :- open(), read(), write(), close()
* **Device** :- ioctl(), read(), write(), close()

b) <u>Layered Structure</u>

user
↓
Application
↓
File system
↓
Device Drivers
↓
Hardware

c) <u>C/System Calls</u>

```
fd = open("file.txt", O_RDONLY);
read (fd, buffer, 100);
close (fd);
```

3) <u>Type of OS</u>

a) Real-time Operating System(RTOS) - gives fast and predictable response for embedded systems.

b) <u>Structure</u>

Application
↓
user Process
↓
Kernel
↓
Hardware

(c) <u>Synchronization System Calls</u>

* sem_wait()
* sem_post()
* pthread_mutex_lock()
* pthread_mutex_unlock()

4) <u>Distributed Operating System</u>

a) Diff:- Distributed OS uses multiple networked computers; Centralized OS uses one central computer.

b) <u>Arch</u>

Node① ←→ Network ←→ Node②

Node① ↕ shared Resources ↕ Network

(c) <u>IPC System Calls</u>

socket() → bind() → listen()
↓
recv() ← send() ← accept()
└→ close()